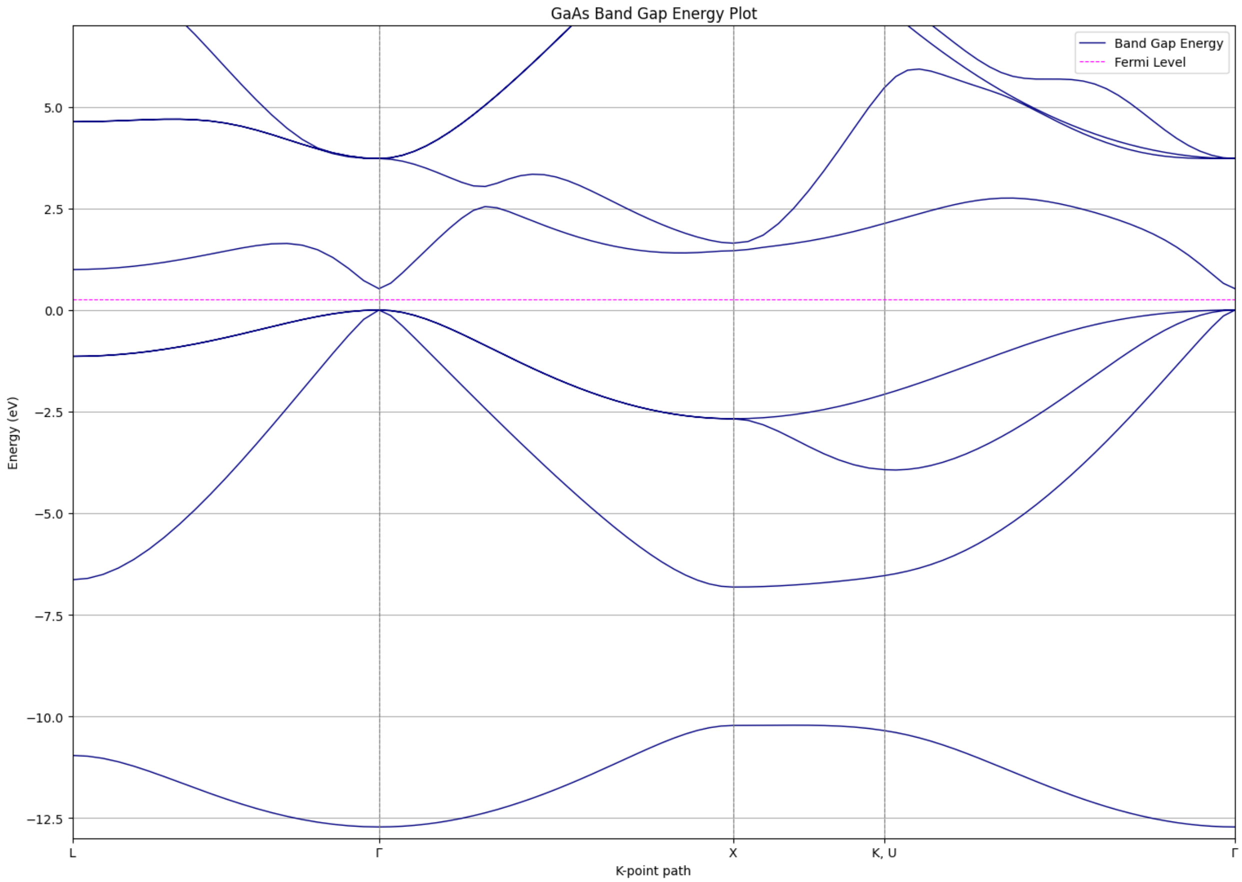
Matthew Athanasopoulos – NE451 Simulation Methods – Lab Assignment 3 – Nov 23rd, 2025

Band Structure Plot (PNG):



Questions (Short Answer):

1) What is the Fermi energy?

From the SCF output, the highest occupied level is 6.7106 eV. For an intrinsic semiconductor, the “Fermi level” customarily used in band plots is the mid-gap, i.e., the midpoint between the valence-band maximum (VBM) and the conduction-band minimum (CBM). Using the extracted VBM/CBM from the NSCF bands, the mid-gap (plotted Fermi reference) is 6.97275 eV, which lies above the VBM as expected.

2) Why coarse k-grid for SCF but a dense path for NSCF?

In the SCF step, the goal is to converge the charge density and total energy, so a reasonably coarse uniform k-grid gives accurate Brillouin-zone integration at much lower cost while the code repeatedly updates the density. In the NSCF bands step, the density is frozen from SCF and we only need eigenvalues along a high-symmetry path (L → Γ → X → K/U → Γ), so instead of a uniform k-grid we sample that path very densely to produce smooth E(k) dispersions without paying the expense of density updates.

3) Is GaAs direct or indirect in this calculation?

Direct. The computed VBM and CBM both occur at Γ, so the gap is direct in the obtained band structure.

4) How were the number of bands chosen?

I set the number of bands to 16 by generally following the assignment's rule of thumb provided. It mentioned to use N_{occc}+ 4–8 bands for an insulator, and N_{occ}+ 8–16 for metals. I chose something on higher end of the insulator rule of thumb (7) since GaAs is a semiconductor.